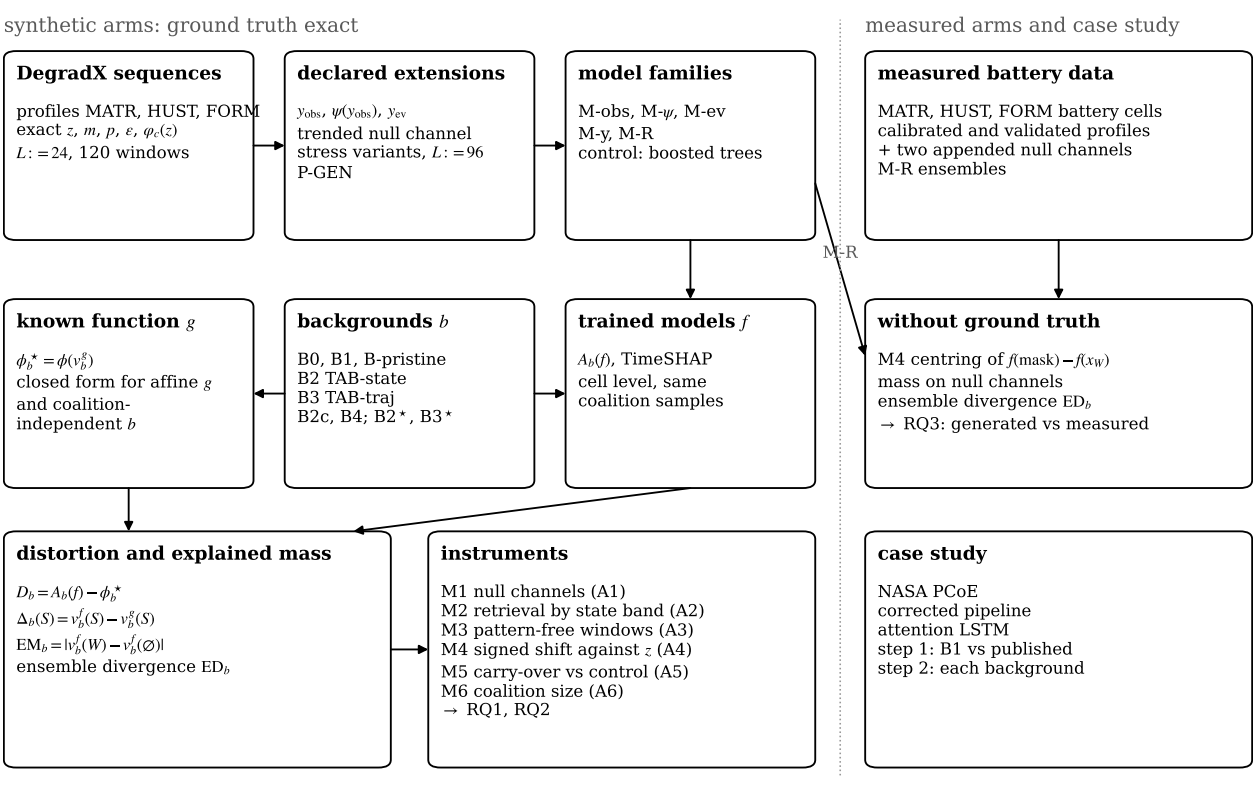
synthetic arms: ground truth exact
measured arms and case study
DegradX sequences
profiles MATR, HUST, FORM
exact z, m, p, ε, $\varphi_c(z)$
$L := 24$, 120 windows
declared extensions
y_{obs}, $\psi(y_{\text{obs}})$, y_{ev}
trended null channel
stress variants, $L := 96$
P-GEN
model families
M-obs, M-ψ, M-ev
M-y, M-R
control: boosted trees
measured battery data
MATR, HUST, FORM battery cells
calibrated and validated profiles
+ two appended null channels
M-R ensembles
M-R
known function g
$\phi_b^\star = \phi(v_b^g)$
closed form for affine g
and coalition-
independent b
backgrounds b
B0, B1, B-pristine
B2 TAB-state
B3 TAB-traj
B2c, B4; B2*, B3*
trained models f
$A_b(f)$, TimeSHAP
cell level, same
coalition samples
without ground truth
M4 centring of $f(\text{mask}) - f(x_W)$
mass on null channels
ensemble divergence ED_b
$\rightarrow$ RQ3: generated vs measured
distortion and explained mass
$D_b = A_b(f) - \phi_b^\star$
$\Delta_b(S) = v_b^f(S) - v_b^g(S)$
$\text{EM}_b = \lvert v_b^f(W) - v_b^f(\varnothing) \rvert$
ensemble divergence ED_b
instruments
M1 null channels (A1)
M2 retrieval by state band (A2)
M3 pattern-free windows (A3)
M4 signed shift against z (A4)
M5 carry-over vs control (A5)
M6 coalition size (A6)
$\rightarrow$ RQ1, RQ2
case study
NASA PCoE
corrected pipeline
attention LSTM
step 1: B1 vs published
step 2: each background